

Samsung's Trade-In & Recycling Programs

A case study in e-waste management and the circular economy

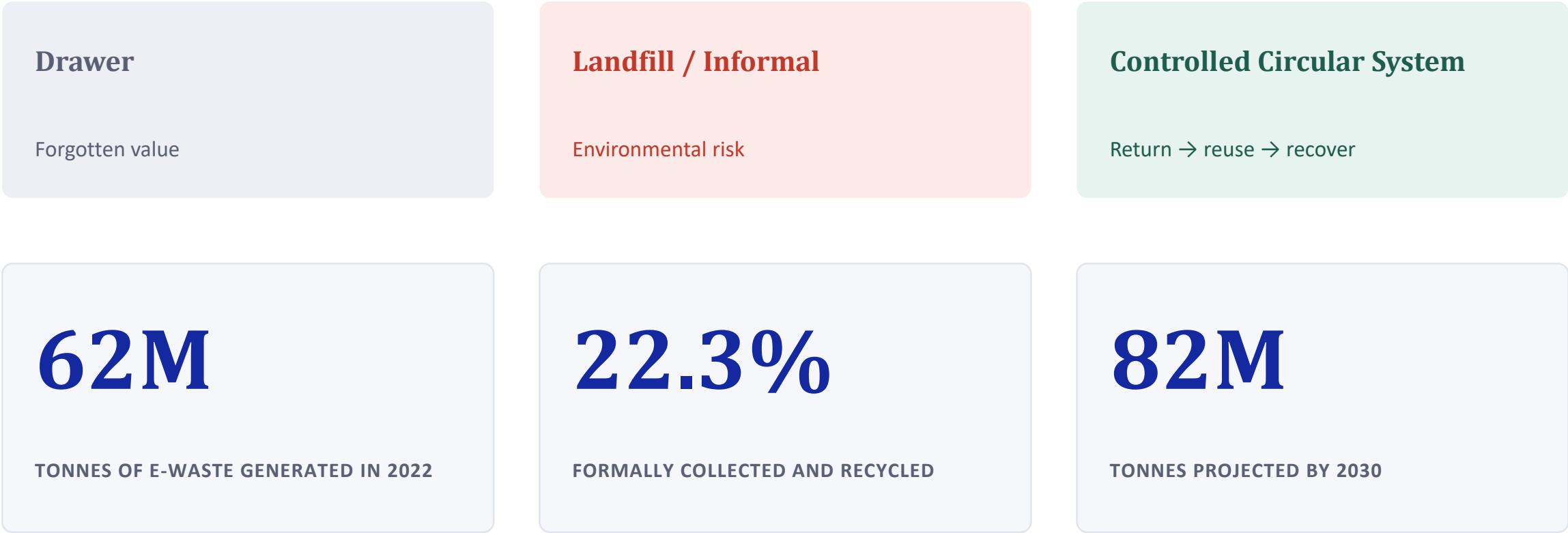
23 SEPTEMBER 2026

PRESENTED BY 10 ENGINEERS



Where does your old smartphone go?

Every smartphone contains both a product with remaining value and materials worth recovering.



Technology is advancing faster than our ability to responsibly manage discarded technology.

A smartphone is not “just waste”

KEY MATERIALS RECOVERED

- Copper
- Gold
- Plastics
- Aluminium
- Cobalt
- Glass

≈31 BILLION KG

of metals contained in 2022's e-waste generation

ENVIRONMENTAL RISKS OF INACTION

- Hazardous substances
- Soil & water contamination
- Unsafe-processing air pollution
- Loss of valuable resources



From “take–make–dispose” to “keep–recover–reuse”

LINEAR

Extract



Manufacture



Buy



Use



Discard

CIRCULAR

Design



Use



Trade-In



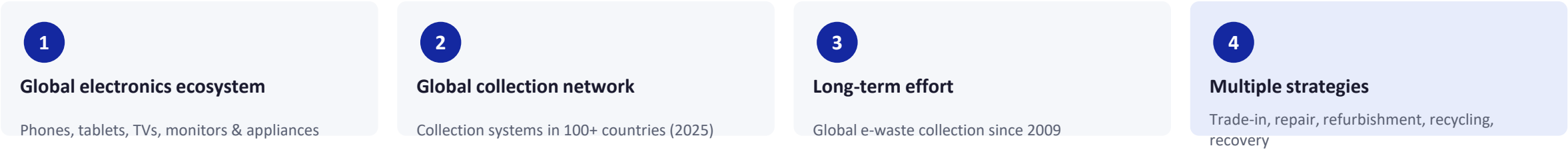
Repair / Refurbish



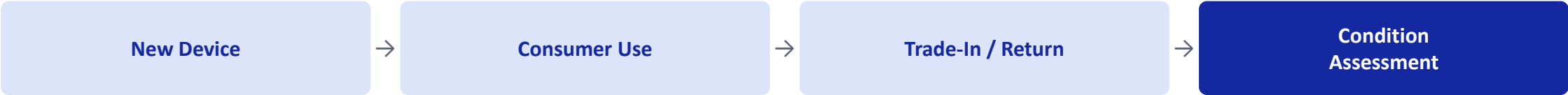
Reuse → Recover → Remanufacture

Keep products and materials at their highest possible value for as long as possible.

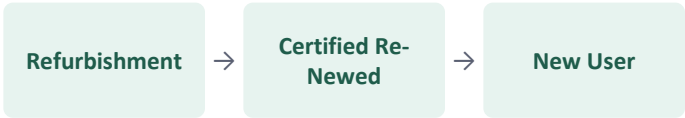
Why Samsung, and how its system works



OPERATING MODEL — NOT EVERY RETURNED DEVICE FOLLOWS THE SAME PATHWAY



REFURBISHMENT PATH

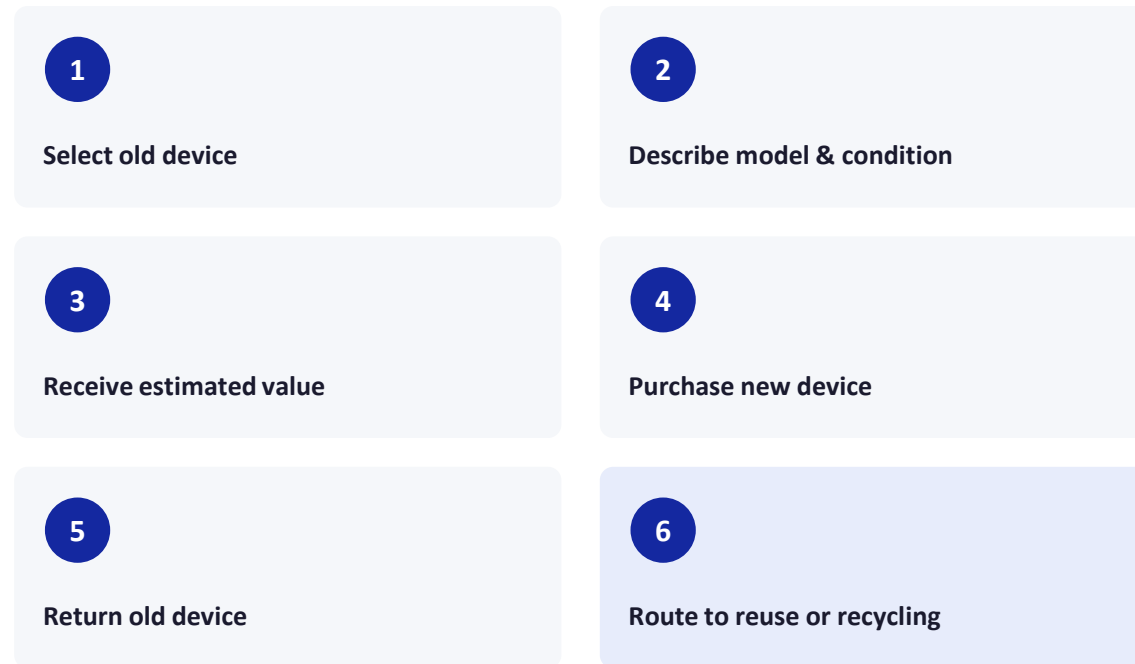


RECYCLING PATH



A useful lens on how a large manufacturer manages products beyond point of sale.

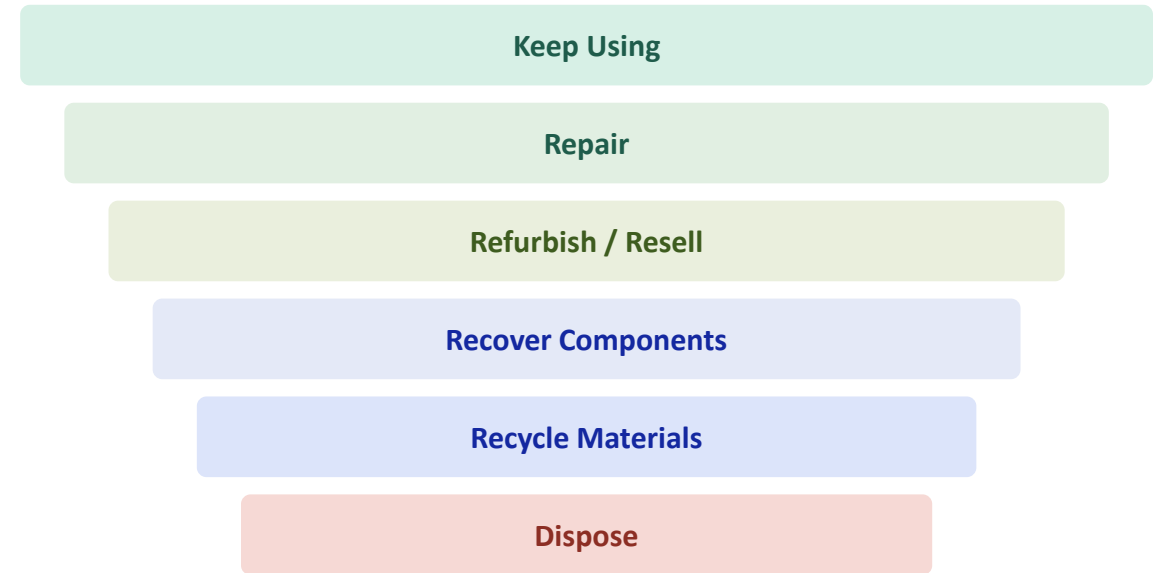
Turning an upgrade into a return



INDIA · Eligible phones, tablets & smartwatches, across multiple manufacturers.

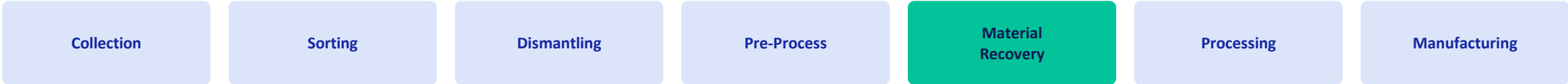
Trade-in is not recycling itself — it is the return mechanism into a controlled circular system.

IS RECYCLING ALWAYS THE BEST FIRST OPTION?

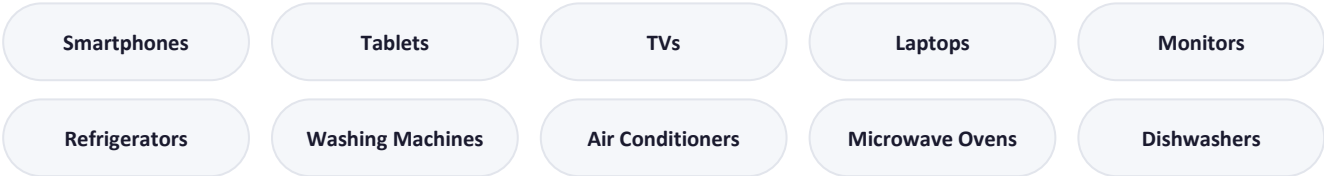


More original value preserved toward the top of the list; refurbishment is not universally best — condition, safety and market context still matter.

Inside the recycling chain



SAMSUNG CARE FOR CLEAN INDIA — WHAT'S COLLECTED



Recycling is not the end of the story — the goal is resource recovery.

Giving devices a second life

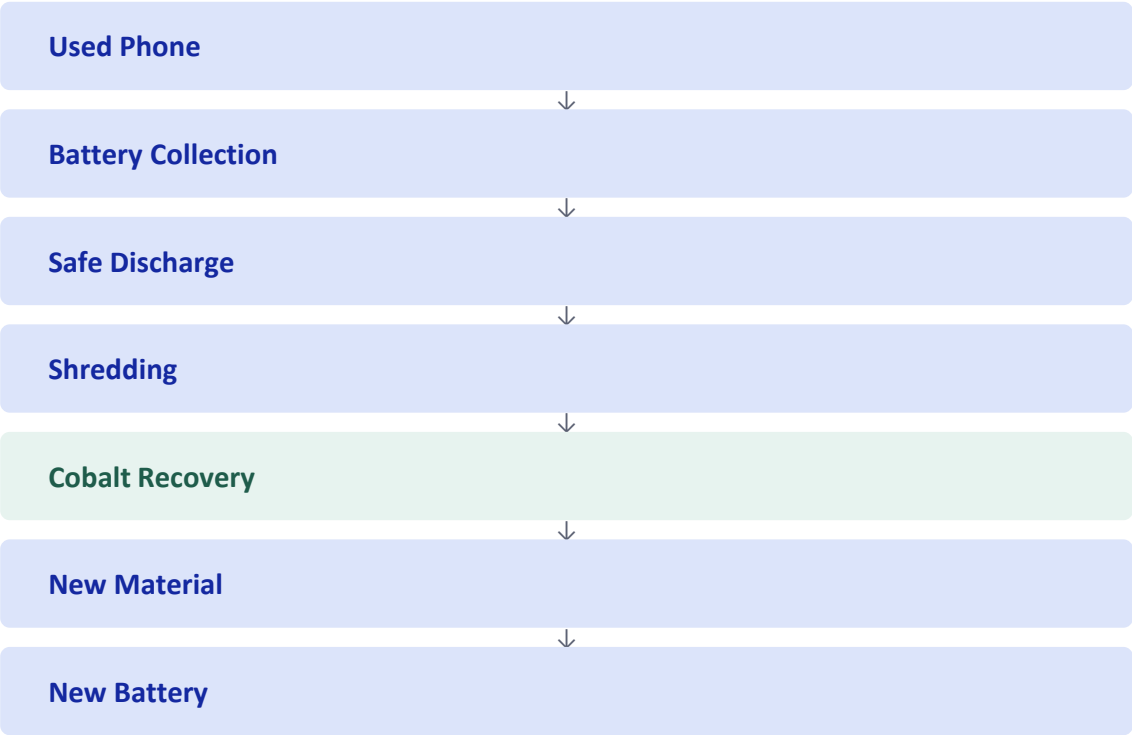


Up to ~85%

of launch price — Galaxy S26 Certified Re-Newed, after inspection, cleaning, refurbishment, testing, certified battery, data reset & software updates, with a 1-year warranty.

Old device → second user → longer product life

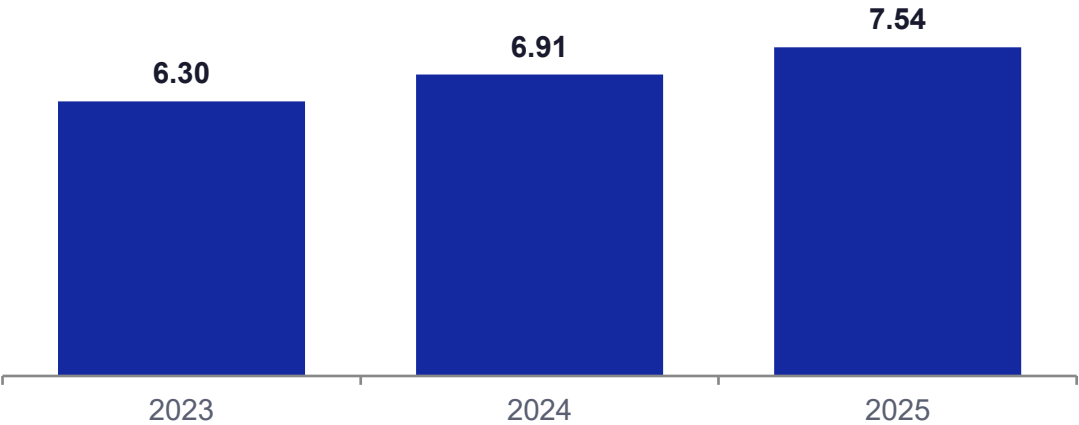
FROM AN OLD GALAXY BATTERY TO A NEW ONE



S24 was first to incorporate recycled cobalt; S25 added advanced recovery from used batteries.

How far it reaches, and where it fits

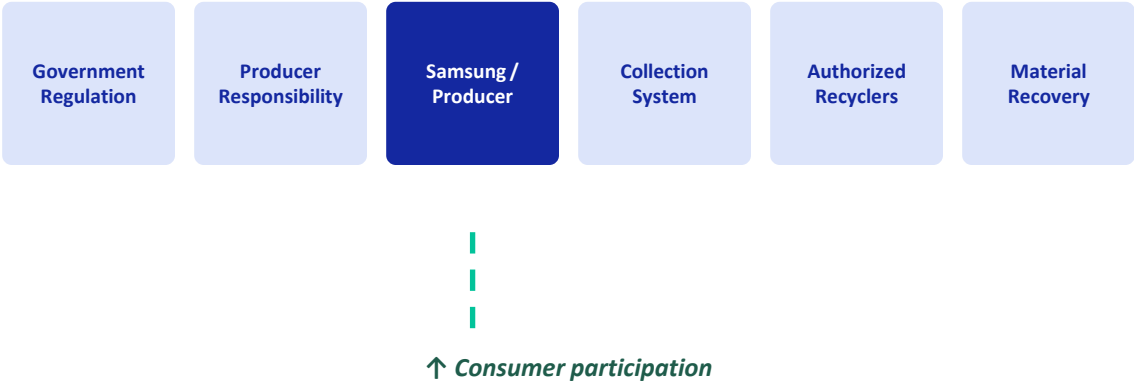
CUMULATIVE E-WASTE COLLECTED, 2009–2025



2009–25 total: 7.54M tonnes • Targets: 10M t by 2030, 25M t by 2050 • Reach: 100+ countries (2025)

Samsung-reported figures.

INDIA'S E-WASTE (MANAGEMENT) RULES, 2022 — EFFECTIVE 1 APR 2023



Samsung operates within a wider regulated system — it does not manage India's entire e-waste system on its own. EPR = producers' obligations for managing post-consumer waste.

Is collection volume enough to measure circularity?

Consumer participation: reluctance to return

Data privacy: personal information concerns

Product complexity: difficult recovery

Infrastructure: country capability varies

Trade-in ≠ reuse: not all devices refurbished

NO — a complete assessment weighs reuse, refurbishment, material recovery and environmental outcomes, not collection volume alone.

1

Extend

Keep products in use longer through repair, trade-in and refurbishment.

2

Recover

Recover components and materials when reuse is impractical.

3

Close the Loop

Return recovered materials to productive use wherever feasible.

The goal is not simply to recycle yesterday's technology — it is to prevent today's technology from becoming tomorrow's waste.